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Parkinson's Disease Modeling in Regenerative Spiny Mice (*Acomys*)

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Protocol status: Working

We use this collection and it's working

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Keywords: disease modeling in regenerative spiny mice, regenerative spiny mice, mechanisms of neurodegeneration, synuclein, parkinson, hallmarks of pd pathology, extensive proteomic alterations in dopaminergic neuron, multifactorial neurodegenerative disorder, key pathological hallmark of pd, neurodegeneration, mice, positive α syn inclusions across multiple brain region, nigrostriatal pathway, mouse α syn pff, robust regeneration of multiple tissue, proteasomal pathway, greater nigrostriatal degeneration, dopaminergic neuron, spiny mouse, relevant pathophysiology across species, regenerative intervention strategies in pd, pd pathology, key pathological hallmark, cell spatial proteomic, pathway, robust regeneration, rodent, neuron, spatial proteomic

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Abstract

Parkinson's disease (PD) is a multifactorial neurodegenerative disorder that has been modeled extensively in animals, primarily rodents, but also in non-human primates and non-mammalian organisms. However, no single animal model fully recapitulates the hallmarks of PD pathology. Here, we extend this work by modeling PD for the first time in the spiny mouse (*Acomys dimidiatus*), a mammal notable for its robust regeneration of multiple tissues. We show that the nigrostriatal pathway of *A. dimidiatus* is vulnerable to both chronic α -synuclein (α Syn) preformed fibril-induced aggregation and acute 6-hydroxydopamine (6-OHDA) toxicity. Mouse α Syn PFFs produced widespread pS129-positive α Syn inclusions across multiple brain regions, mirroring a key pathological hallmark of PD. Compared to C57BL/6J mice, *A. dimidiatus* exhibited more pronounced behavioral impairments, greater nigrostriatal degeneration, and a higher pS129- α Syn inclusion burden within substantia nigra pars compacta (SNpc) neurons. To probe the molecular underpinnings behind the vulnerability, we performed single-cell spatial proteomics, which revealed extensive proteomic alterations in dopaminergic neurons associated with α Syn aggregation. While proteasomal pathways emerged as major PD-related hubs conserved in both species, multiple proteins and pathways were uniquely differentially dysregulated between the two species. Notably, proteomic analysis identified heightened astrocytic activation in the SNpc, which we validated histologically, suggesting a distinct glial response compared to mice. Together, these findings expand our understanding of PD-relevant pathophysiology across species and establish *A. dimidiatus* as a model for studying mechanisms of neurodegeneration and regenerative intervention strategies in PD. PNAS may not allow for final submission (similar ones highlighted in the later sections)

Files

 SEARCH

Protocol

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scSP: Single Cell Spatial Proteomics LC-MS/MS Analysis of LCM-isolated single-cell mouse brain tissue

VERSION 2

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Spiny mouse (*Acomys*) stereotaxic injection

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